Vedanta Life Guidance – An AI-Powered Advaita Vedanta Assistant

Chapter 1: Introduction

Vedanta Life Guidance is an AI-powered conversational application designed to help individuals explore practical perspectives on everyday life challenges through the teachings of Advaita Vedanta.

In today's fast-paced world, individuals often encounter challenges related to stress, anger, relationships, career decisions, fear, attachment and personal identity. Philosophical teachings can offer valuable perspectives for understanding these experiences and developing greater self-awareness. However, accessing and connecting traditional philosophical texts with modern-day situations can sometimes be difficult.

This project aims to bridge that gap by combining the principles of Advaita Vedanta with modern Artificial Intelligence technologies. The application uses a Retrieval-Augmented Generation (RAG) approach to retrieve relevant passages from a curated knowledge base of verified philosophical texts and uses a Large Language Model (LLM) to generate contextual responses based on the retrieved information.

The initial knowledge base is developed using The Essential Vedanta: A New Source Book of Advaita Vedanta, edited by Eliot Deutsch and Rohit Dalvi. Selected passages are organised with their original references, explanations, keywords and related practical situations to support accurate information retrieval and source-based responses.

Users can interact with the application through a conversational interface in English or Hindi. Based on the user's question, the system retrieves relevant philosophical teachings and presents an explanation along with practical reflections. The application also supports conversational context through session-based conversation history, allowing users to ask follow-up questions within the same conversation.

Purpose of Project

The application aims to:

- Make selected Advaita Vedanta teachings easier to understand.
- Connect philosophical concepts with common life situations.
- Provide source-based explanations with relevant references.
- Support interaction in both English and Hindi.
- Encourage self-reflection through AI-generated practical perspectives.

Technology Overview

Component	Technology	Purpose
Frontend	React.js	Develops the interactive user interface and conversational chat functionality.
Backend	FastAPI	Handles API requests, application logic and communication between frontend and AI components.
Large Language Model (LLM)	Qwen3:4b	Generates contextual responses based on retrieved philosophical passages.
Model Runtime	Ollama	Runs and manages the LLM locally.
LLM Framework	LangChain	Manages prompts, model interactions and response generation.
Embedding Model	BGE-M3	Converts text passages and user queries into multilingual vector embeddings.
Vector Database	Qdrant	Stores vector embeddings and retrieves relevant passages using similarity search.
Knowledge Base	JSON	Stores selected Vedanta passages, explanations, keywords and source metadata.
Conversation Management	Python In-memory Storage	Maintains session-based conversation history for follow-up questions.
Programming Language	Python	Implements backend logic, retrieval functionality and AI integration.
API Communication	Axios	Enables communication between the React frontend and FastAPI backend.

Project Scope

The current implementation focuses on developing a working prototype that demonstrates how AI and Retrieval-Augmented Generation can be used to make philosophical knowledge accessible through a conversational application.

The scope includes:
- Preparing a structured knowledge base of 15–20 selected passages from the provided source PDF.
- Generating multilingual embeddings for the selected passages.
- Storing and retrieving passages using Qdrant.
- Integrating a locally running Large Language Model through Ollama.
- Generating responses grounded in retrieved philosophical content.
- Providing relevant source references with generated responses.
- Supporting English and Hindi interactions.
- Maintaining conversational context through session-based history.
- Handling unrelated questions without forcing an unsupported philosophical answer.

Chapter 2: Problem Statement

Background

Advaita Vedanta is a philosophical tradition that explores the nature of the Self (Ātman), ultimate reality (Brahman) and the relationship between individual identity and the world.

Its teachings are available through various traditional texts, including the Upanishads and the Bhagavad Gita. However, understanding these philosophical concepts and connecting them with modern everyday situations can be challenging for individuals who are unfamiliar with the subject.

People commonly experience challenges related to stress, anger, fear, relationships, career decisions, attachment and personal identity. Although philosophical teachings can provide perspectives for reflecting on these experiences, accessing relevant passages and understanding their practical significance often requires familiarity with traditional literature and its terminology.

Proposed Problem Solution

To address these challenges, the project proposes an AI-powered conversational assistant based on Retrieval-Augmented Generation (RAG).

The application maintains a curated knowledge base of selected Advaita Vedanta passages. When a user submits a question, the system converts the query into an embedding and retrieves semantically relevant passages from Qdrant.

The retrieved passages are then provided as context to a Large Language Model, which generates an explanation and practical reflection based on the available source material.

The application also provides source references to help users identify the philosophical texts supporting the response. It supports English and Hindi interactions and maintains session-based conversation history to enable contextual follow-up questions.

For questions unrelated to the available knowledge base, the system is designed to avoid generating unsupported Vedanta-based explanations.

2.4 Expected Outcome

The expected outcome is a working AI-powered application that makes selected Advaita Vedanta teachings easier to access, understand and explore through natural-language conversations.

The system aims to provide a connection between traditional philosophical knowledge and contemporary life situations while maintaining source traceability and distinguishing source-based explanations from AI-generated practical reflections.

Chapter 3: Project Objectives

The specific objectives of the project are as follows:

3.1 Development of a Curated Knowledge Base
- Select 15–20 relevant passages from the provided book, *The Essential Vedanta: A New Source Book of Advaita Vedanta*.
- Organise the selected passages into structured JSON records.
- Include source references, author or commentator details, page numbers, keywords and simplified explanations.
- Prepare English and Hindi explanations to support multilingual interaction.

3.2 Implementation of Retrieval-Augmented Generation (RAG)
- Generate multilingual vector embeddings using the BGE-M3 embedding model.
- Store the embeddings and associated metadata in Qdrant.
- Retrieve relevant philosophical passages based on the semantic meaning of user questions.
- Provide retrieved passages as context to the Large Language Model to generate grounded responses.

3.3 Integration of Artificial Intelligence
- Integrate the Qwen3:4b Large Language Model through Ollama.
- Use LangChain to manage prompts and interactions with the language model.
- Generate contextual explanations based on retrieved knowledge.

3.4 Development of a Conversational User Interface
- Develop an interactive frontend using React.js.
- Allow users to submit questions related to everyday life challenges.
- Support English and Hindi conversations.
- Display generated answers along with relevant philosophical source references.
- Provide loading indicators and error handling for a better user experience.

3.5 Session-Based Conversation Management
- Generate unique session IDs for individual conversations.
- Maintain conversation history to support contextual follow-up questions.
- Allow users to start a new conversation independently of previous sessions.
- Pass relevant conversation history to the language model for improved conversational continuity.

.6 Source-Based and Responsible Responses
- Provide relevant source references alongside generated explanations.
- Distinguish philosophical teachings from AI-generated practical reflections.
- Avoid forcing answers when relevant supporting passages are unavailable.
- Present the application as an educational and philosophical exploration tool rather than a replacement for professional guidance.

Chapter 4: Proposed Solution

Working Approach

The application follows the workflow described below:

Step 1: User Interaction
The user accesses the React-based interface and submits a question related to a personal or everyday life situation, such as managing anger, handling stress, making decisions or understanding attachment.

Step 2: Request Handling
The question is sent from the React frontend to the FastAPI backend through an API request. A unique session ID is also included to identify the current conversation and maintain its context.

Step 3: Query Embedding
The BGE-M3 embedding model converts the user's question into a numerical vector representation that captures its semantic meaning.

Step 4: Relevant Passage Retrieval
The generated query embedding is compared against the stored passage embeddings in Qdrant. The retriever identifies passages that are semantically relevant to the user's question.

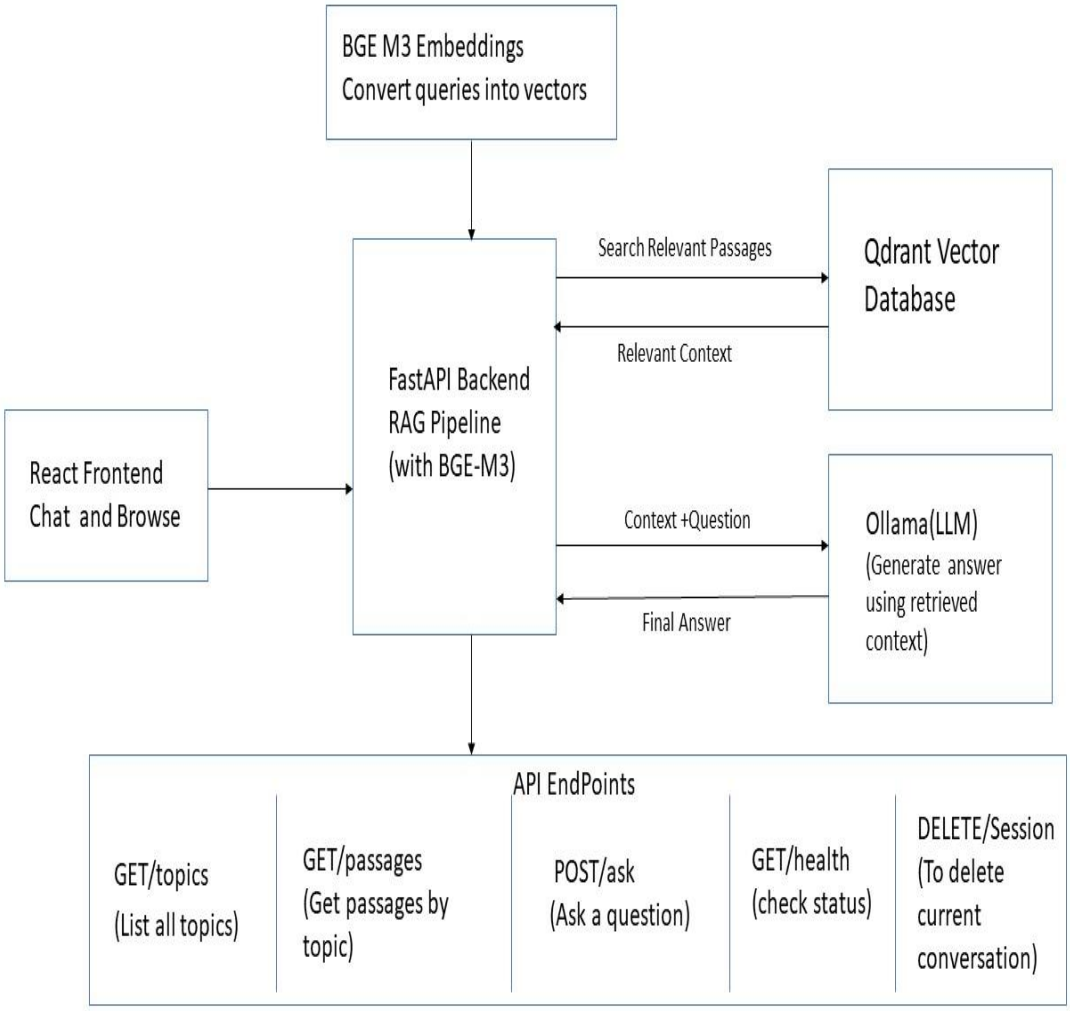
Step 5: Context Preparation
The retrieved passages and their associated metadata are prepared as contextual information. Relevant conversation history is also supplied to help the language model understand follow-up questions.

Step 6: Response Generation
The retrieved context is passed to the Qwen3:4b language model running through Ollama. LangChain manages the prompt and model interaction. The model generates a response based on the available philosophical material and the conversation context.

Step 7: Response Delivery
The generated response and relevant source references are returned to the React frontend. The user can view the explanation and continue the conversation through follow-up questions.

Step 8: Conversation History Management
The user's question and the assistant's response are stored in an in-memory conversation store using the session ID. This allows subsequent questions within the same session to use previous conversational context.

Chapter 5: System Architecture



The system follows these steps when a user submits a question:

1. The user enters a question through the React frontend.
2. The frontend sends the question and session ID to the FastAPI backend.
3. The backend uses BGE-M3 to convert the question into an embedding.
4. The embedding is used to search the Qdrant vector database for relevant Vedanta passages.
5. Qdrant returns the matching passages and their associated information to the backend.
6. The backend combines the original question with the retrieved context and sends it to the Ollama-hosted language model.
7. The language model generates a response using the retrieved context.
8. The backend returns the final answer and relevant source information to the React frontend.
9. The conversation history is maintained using session-based storage so that follow-up questions can use previous conversation context.

API EndPoints

API Endpoint	HTTP Method	Description
`/topics`	GET	Retrieves the list of all available topics in the knowledge base.
`/passages`	GET	Retrieves passages from the knowledge base, with an optional topic filter.
`/ask`	POST	Accepts a user's question and session ID, then returns the generated answer and relevant source passages.
`/health`	GET	Checks the health status of the backend application.
`/sessions/{session_id}`	DELETE	Deletes the conversation history associated with the specified session ID.

Chapter 6: Testing and Evaluation

Introduction

Testing was performed to evaluate the functionality and reliability of the Vedanta Life Guidance application. The main objective was to verify that users could submit questions, retrieve relevant passages from the Vedanta knowledge base, receive context-based responses, and continue conversations through session management.

The testing process focused on the frontend, backend APIs, semantic retrieval, language model integration, bilingual query handling, and the application's response to questions outside its knowledge domain.

Testing Objectives

The primary objectives of testing were:

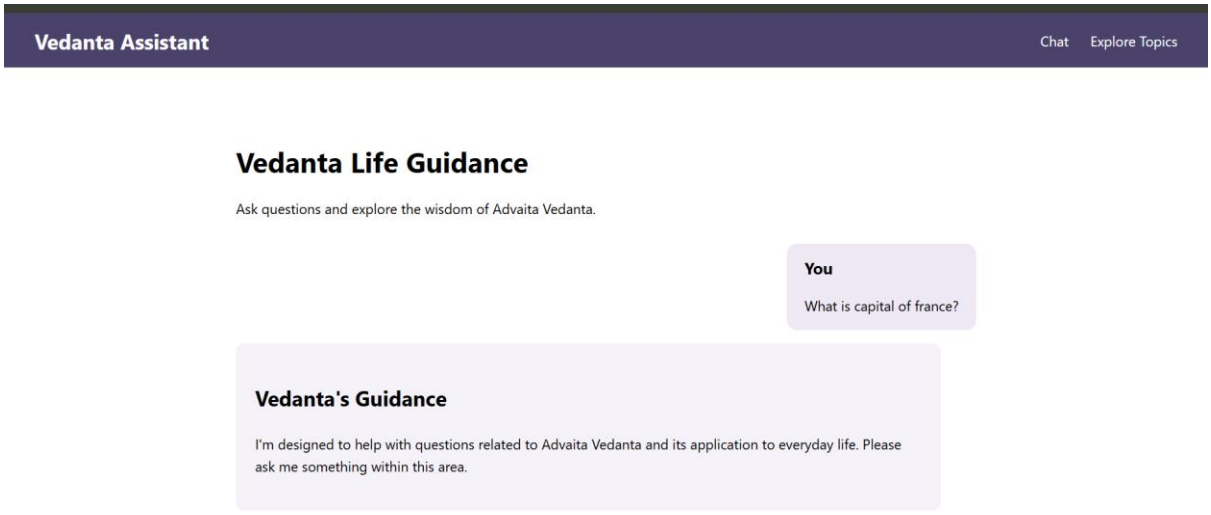
- To verify communication between the React frontend and FastAPI backend.
- To check whether user questions are converted into embeddings correctly.
- To evaluate the retrieval of relevant passages from the Qdrant vector database.
- To verify that the language model generates responses using the retrieved context.
- To test English and Hindi question handling.
- To check whether conversation history supports follow-up questions.
- To evaluate how the application handles unrelated or unsupported questions.

Test Cases

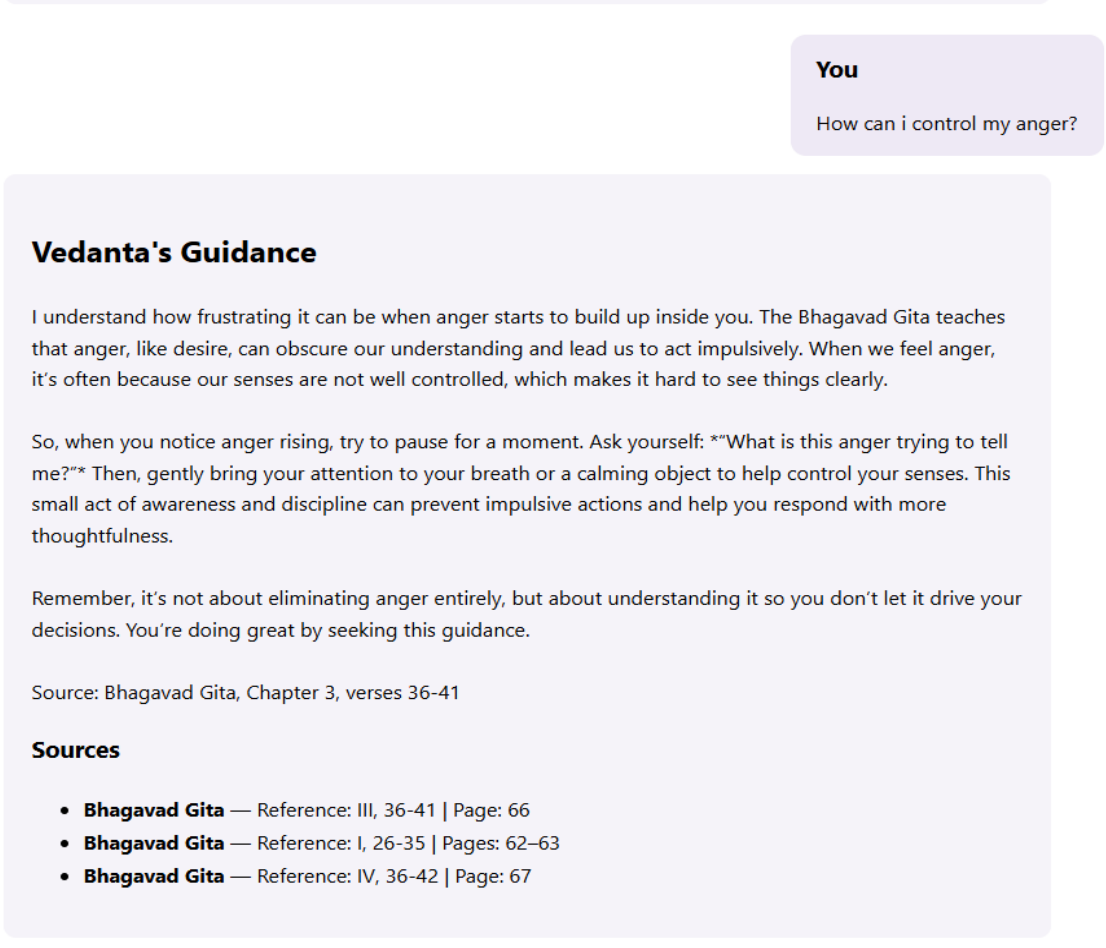
TestCase ID	Scenario	Test Input	Expected Result	Actual Result	Status
01	Backend health check	Send a GET request to /health.	Backend returns its health status.	Backend returned its health status as ok	Pass
02	Retrieve topics	Send a GET request to /topics.	List of available knowledge-base topics should return.	List of available knowledge-base topics is returned.	Pass
03	Retrieve passages	Send a GET request to /passages.	Passages from the knowledge base should returned.	Passages from the knowledge base are returned.	Pass
04	English Question	How can i control my anger?	Relevant Vedanta passages should be retrieved and used to generate a response.	Relevant Vedanta passages are retrieved and generated a response.	Pass
05	Follow up Question	How can I apply this in daily life	Previous conversation context should be considered while processing the follow-up.	Previous conversation context is considered while processing the follow-up.	Pass
06	New Converstion	Start a new conversation using the New Conversation button.	A new session should be created and the previous conversation should not carried into it.	A new session is created and the previous conversation is not carried into it.	Pass
07	Unsupported Question	What is capital of france	The application handles the unrelated question without presenting unsupported information as Vedanta-based guidance.	The application handles the unrelated question without presenting unsupported information as Vedanta-based guidance.	Pass

Testing Questions

Unsupported Question



Supported Question



Chapter 7: Future Scope and Scalability

The current Vedanta Life Guidance application demonstrates how Retrieval-Augmented Generation can be used to connect the teachings of Advaita Vedanta with everyday life questions. Although the prototype provides basic conversational functionality, semantic retrieval, and AI-generated responses, several improvements can be introduced to make the application more efficient, reliable, and scalable.

Future enhancements can focus on improving the knowledge base, conversation management, response generation, user experience, and deployment infrastructure.

Future Enhancements

1. Expansion of the Knowledge Base

The current knowledge base contains selected passages from The Essential Vedanta: A New Source Book of Advaita Vedanta. In the future, it can be expanded with additional passages and explanations from other authentic Vedanta texts. This would improve the coverage of philosophical concepts and allow the assistant to address a wider range of questions.

2. Persistent Conversation Storage

The current implementation uses in-memory storage to maintain conversation history. This data is lost when the backend restarts. In the future, Redis or a relational database such as PostgreSQL can be integrated to maintain conversation history persistently.

3. User Authentication

User registration and login functionality can be introduced to provide personalized experiences. Authenticated users could maintain their conversation history across multiple sessions and access previously saved conversations.

4. Improved Retrieval Accuracy

The retrieval pipeline can be improved through better passage organization, metadata filtering, hybrid search, and retrieval evaluation. These improvements could help the application identify more relevant passages for complex or context-dependent questions.

5. Performance Optimization

The current application runs the language model locally through Ollama. Future deployment could use GPU-based inference or optimized model-serving infrastructure to reduce response generation time and support more simultaneous requests.

6. Mobile Application Support

A mobile application could be developed using technologies such as React Native or Flutter. This would make Vedanta Life Guidance accessible to users through smartphones and tablets.

Scalability Strategy

Scalability refers to the ability of an application to handle an increasing number of users and requests without significant degradation in performance.

The current prototype is designed primarily for demonstration and limited usage. To support a larger user base, the following improvements can be introduced.

Scalability at Different User Levels

The following table presents a possible future infrastructure strategy for different application usage levels. These are proposed deployment approaches, not measured capacity results.

User Scale	Proposed Infrastructure	Purpose
Around 1,000 users	Cloud-hosted FastAPI, Redis, managed database, Qdrant, and suitable model-serving infrastructure.	Support a growing user base with centralized storage and improved availability.
Around 10,000 users	Multiple FastAPI instances, load balancer, caching, dedicated GPU inference, and optimized database services.	Distribute requests and improve response generation performance.
Around 100,000 users	Horizontally scaled backend services, distributed vector database, GPU inference cluster, request queues, and autoscaling.	Support high request volumes and maintain service availability.

Actual infrastructure requirements would depend on concurrent users, request frequency, model size, response length, and available computing resources.

Authentication and Session Management in the Future

The current application identifies conversations using unique session IDs and maintains their history in backend memory.

In a future version, user authentication can be introduced using secure password hashing and access and refresh tokens. Each authenticated user could have separate conversation records stored in a persistent database.

A logout operation could revoke the user's authentication session or refresh token and clear the authentication data from the frontend. Conversation history could be retained or deleted according to the application's data-retention policy.

This would provide better user management, privacy, and continuity across multiple visits.

Summary

The current Vedanta Life Guidance prototype establishes the foundation for an AI-powered philosophical guidance application. Future improvements can focus on persistent storage, authentication, retrieval accuracy, performance optimization, and mobile accessibility.